Black rays $\mathcal{C}$ and $\mathcal{D}$ in Figure 8.2 begin on the horizon, travel outward and get trapped on nonequatorial orbits in the shell of fire, and then escape from their trapped orbits and come to your eyes, bringing images of bits of the shadow edge that lie outside the equatorial plane. The trapped orbit for ray $\mathcal{D}$ is shown in the upper right inset. White rays $\mathcal{C}$ and $\mathcal{D}$ (not shown), coming from distant stars, get trapped alongside black rays $\mathcal{C}$ and $\mathcal{D}$, and then travel to your eyes alongside $\mathcal{C}$ and $\mathcal{D}$, bringing images of bits of the ring of fire alongside bits of the shadow edge.

Lensing by a Nonspinning Black Hole

To understand the pattern of gravitationally lensed stars outside the shadow and their streaming as the camera moves, let's begin with a nonspinning black hole and with light rays that emerge from a single star (Figure 8.3). Two light rays travel from the star to the camera. They each travel along the straightest line they can in the hole's warped space, but because of the warping, each ray gets bent.

One bent ray travels to the camera around the hole's left side; the other, around its right side. Each ray brings the camera its own image of the star. The two images, as seen by the camera, are shown in the inset of Figure 8.3. I put red circles around them to distinguish them from all the other stars the camera sees. Note that the right image is much closer to the hole's shadow than the left image. This is because its bent ray passed closer to the hole's event horizon.